# STA 310: Homework 1

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```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

## Instructions

- Write all narrative using full sentences. Write all interpretations and conclusions in the context of the data.
- Be sure all analysis code is displayed in the rendered pdf.
- If you are fitting a model, display the model output in a neatly formatted table. (The `tidy` and `kable` functions can help!)
- If you are creating a plot, use clear and informative labels and titles.
- Render and back up your work reguarly, such as using Github.
- When you're done, we should be able to render the final version of the Rmd document to fully reproduce your pdf.
- Upload your pdf to Gradescope. Upload your Rmd, pdf (and any data) to Canvas.

## Exercises

Exercises 1 - 4 are adapted from exercises in Section 1.8 of @roback2021beyond.

### Exercise 1

Consider the following scenario:

> Researchers record the number of cricket chirps per minute and temperature during that time. They use linear regression to investigate whether the number of chirps varies with temperature.

a. Identify the response and predictor variable.

   The response variable in this case would be the number of chirps, and the predictor would be the temperature.

b. Write the complete specification of the statistical model.

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

$$i = 1, .., n$$

Y_i = number of chirps per minute at the ith observation

X_i = temperature at the ith observation

B_0 = the expected chirps per minute when the temperature is 0

B_1 = the expected increase in chirps per minute for every one unit increase in temperature

e_i = random normal error term

a. Write the assumptions for linear regression in the context of the problem.

   Normality- The data are approximately normally distributed

   Independence - Chirp counts from different minutes are counted for and not repeated.

   Constant Variance - The variability of is the same across all temperatures.

   Linearity - There is a linear relationship between the mean response and the predictor variables.

## Exercise 2

Consider the following scenario:

> A randomized clinical trial investigated postnatal depression and the use of an estrogen patch. Patients were randomly assigned to either use the patch or not. Depression scores were recorded on 6 different visits.

a. Identify the response and predictor variables.

   The response variable in this case would be the depression score. The predictor variables are the treatment group (estrogen patch or not) and visits.

b. Identify which model assumption(s) are violated. Briefly explain your choice.

   The model assumption that is violated is the assumption of independence because the same patients were recorded for the same thing on 6 different occasions therefore the patients are correlated.

## Exercise 3

Use the Kentucky Derby case study in Chapter 1 of *Beyond Multiple Linear Regression.*

a. Consider Equation (1.3) in Section 1.6.3. Show why we have to be sure to say "holding year constant", "after adjusting for year", or an equivalent statement, when interpreting $\beta_2$.

   **We have to say holding year constant because**

$$\beta_2$$

   **represents the difference in expected win speed between fast and non fast track conditions for the same year. Since the variables year and fast are correlated, not saying "holding year constant" would incorrectly attribute improvements over time to the effect of the fast track conditions.**

b. Briefly explain why there is no error (random variation) term $\epsilon_i$ in Equation (1.4) in Section 1.6.6?

```
There is no error term in the equation because it represents the fitted values
from the model, and not the random data generating process. The random error
term is apart of the true model for Y_i but it is dropped when writing the
deterministic equation for the estimated mean response.
```

## Exercise 4

**The data set `kingCountyHouses.csv` in the `data` folder contains data on over 20,000 houses sold in King County, Washington (@kingcounty).**

**We will use the following variables:**

- **`price` = selling price of the house**
- **`sqft` = interior square footage**

*See Section 1.8 of Beyond Multiple Linear Regression for the full list of variables.*

a. **Fit a linear regression model with `price` as the response variable and `sqft` as the predictor variable (Model 1). Interpret the slope coefficient in terms of the expected change in price when `sqft` increases by 100.**

```r
library(readr)
kingCountyHouses <-
read_csv("~/STA310/generalized-linear-models/homeworks/data/kingCountyHouses.csv")
```

```
## Rows: 21613 Columns: 9
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## dbl (9): price, date, bedrooms, bathrooms, sqft, floors, waterfront, yr_buil...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
model1 <- lm(price ~ sqft, data = kingCountyHouses)
summary(model1)
```

```
##
## Call:
## lm(formula = price ~ sqft, data = kingCountyHouses)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1476062  -147486   -24043   106182  4362067
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -43580.743   4402.690  -9.899   <2e-16 ***
## sqft           280.624      1.936 144.920   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 261500 on 21611 degrees of freedom
## Multiple R-squared:  0.4929, Adjusted R-squared:  0.4928
## F-statistic: 2.1e+04 on 1 and 21611 DF,  p-value: < 2.2e-16
```

**Increasing the size of a house by 100 square feet is associated with an average increase in selling price of about $28,000 while holding everything else constant.**

b. Fit Model 2, where `logprice` (the natural log of price) is now the response variable and `sqft` is still the predictor variable. How is the `logprice` expected to change when `sqft` increases by 100?

```r
model2 <- lm(log(price) ~ sqft, data = kingCountyHouses)
summary(model2)
```

```
##
```

```
## Call:
## lm(formula = log(price) ~ sqft, data = kingCountyHouses)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.97781 -0.28543  0.01472  0.26070  1.27628
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 1.222e+01  6.374e-03  1916.9   <2e-16 ***
## sqft        3.987e-04  2.803e-06   142.2   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3785 on 21611 degrees of freedom
## Multiple R-squared:  0.4835, Adjusted R-squared:  0.4835
## F-statistic: 2.023e+04 on 1 and 21611 DF,  p-value: < 2.2e-16
```

**When the interior square footage of a home increases by 100 square feet, the expected value of logprice increases by 0.0399 (~4% increase in price for a 100 square foot increase in size).**

c. **Recall that $log(a) - log(b) = log(\frac{a}{b})$. Use this to derive how the `price` is expected to change when `sqft` increases by 100 based on Model 2.**

**From model2**

$$log(price) = \beta_0 + \beta_1 sqft$$

**, if sqft increases by 100 the change in logprice is**

$$\Delta log(price) = 100\beta_1$$

**. Log identity =**

$$log(price_n) - log(price_o) = log(price_n/log(price_o)) = 100\beta_1$$

**. Where**

$$\beta_1 = 0.0003987$$

,

$$log(price_n)/log(price_o) = 0.0399$$

. Exponentiating =

$$price_n/price_o =$$

e^0.0399 = 1.041. **Thus, when squarefeet increases by 100, the expected price increases by a factor of about 1.041 , ~4.1% increase in price.**

d. Fit Model 3, where `price` and `logsqft` (the natural log of sqft) are the response and predictor variables, respectively. How does the price expected to change when sqft increases by 10%? *As a hint, this is the same as multiplying sqft by 1.10.*

```r
model3 <- lm(price ~ log(sqft), data = kingCountyHouses)
summary(model3)
```

```
##
## Call:
## lm(formula = price ~ log(sqft), data = kingCountyHouses)
##
## Residuals:
```

```
##     Min      1Q  Median      3Q     Max
## -606252 -170067  -33139  106342 6183772
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -3451377      35169  -98.14   <2e-16 ***
## log(sqft)     528648       4651  113.67   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 290400 on 21611 degrees of freedom
## Multiple R-squared:  0.3742, Adjusted R-squared:  0.3742
## F-statistic: 1.292e+04 on 1 and 21611 DF,  p-value: < 2.2e-16
```

**When interior square footage increases by 10%, the expected selling price increases by approximately $50,000, on average.**

Click here for notes on interpreting model effects for log-transformed response and/or predictor variables.

## Exercise 5

The goal of this analysis is to use characteristics of 593 colleges and universities in the United States to understand variability in the early career pay, defined as the median salary for alumni with 0 - 5 years of experience. The data was obtained from TidyTuesday College tuition, diversity, and pay, and was originaly collected from the PayScale College Salary Report.

The data set is located in `college-data.csv` in the `data` folder. We will focus on the following variables:

| variable | class | description |
|---|---|---|
| name | character | Name of school |
| state_name | character | state name |
| type | character | Public or private |
| early_career_pay | double | Median salary for alumni with 0 - 5 years experience (in US dollars) |
| stem_percent | double | Percent of degrees awarded in science, technology, engineering, or math subjects |
| out_of_state_total | double | Total cost for in-state residents in USD (sum of room & board + out of state tuition) |

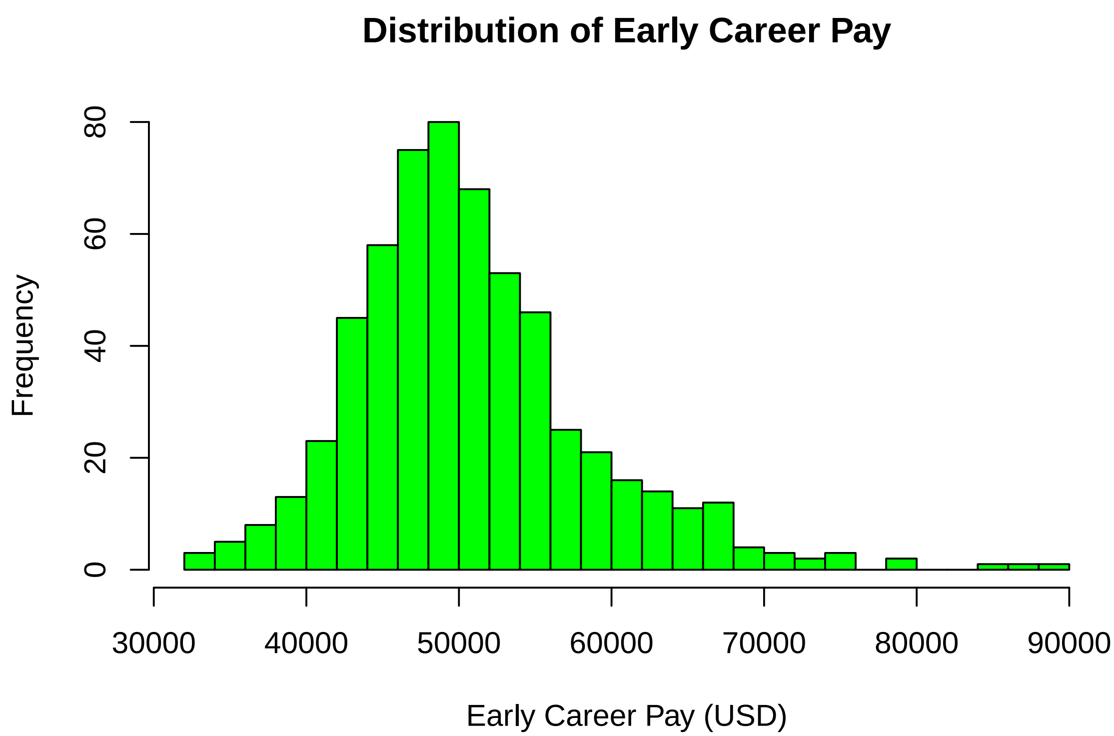
a. Visualize the distribution of the response variable `early_career_pay`. Write 1 - 2 observations from the plot.

```r
college_data <- read_csv("~/STA310/generalized-linear-models/homeworks/data/college-data.csv")
```

```
## Rows: 593 Columns: 15
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (5): name, state, state_code, type, degree_length
## dbl (10): room_and_board, in_state_tuition, in_state_total, out_of_state_tui...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
hist(college_data$early_career_pay,
     main = "Distribution of Early Career Pay",
     xlab = "Early Career Pay (USD)",
```

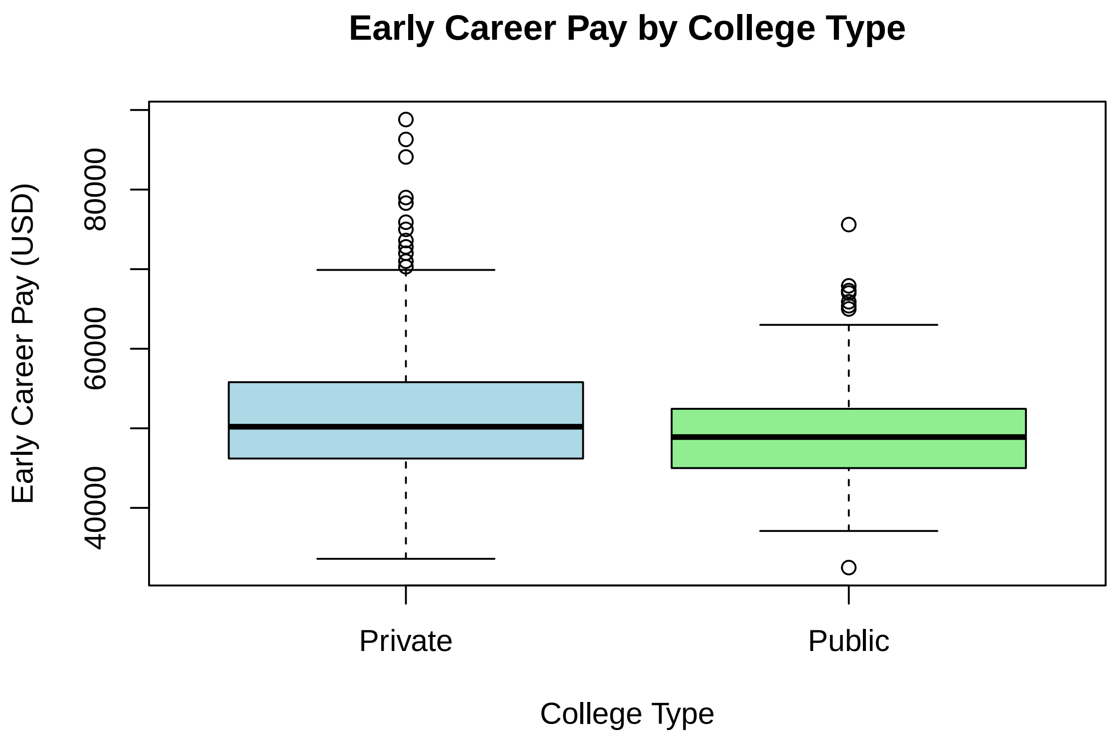
```r
     ylab = "Frequency",
     col = "green",
     breaks = 30)
```




**Some observations of this plot are that the most common early carer pay lie between the range of about $47,000 to about $52,000. The data appear to be right skewed as there are very few values higher than $60,000 in comparison to the values lower than $60,000.**

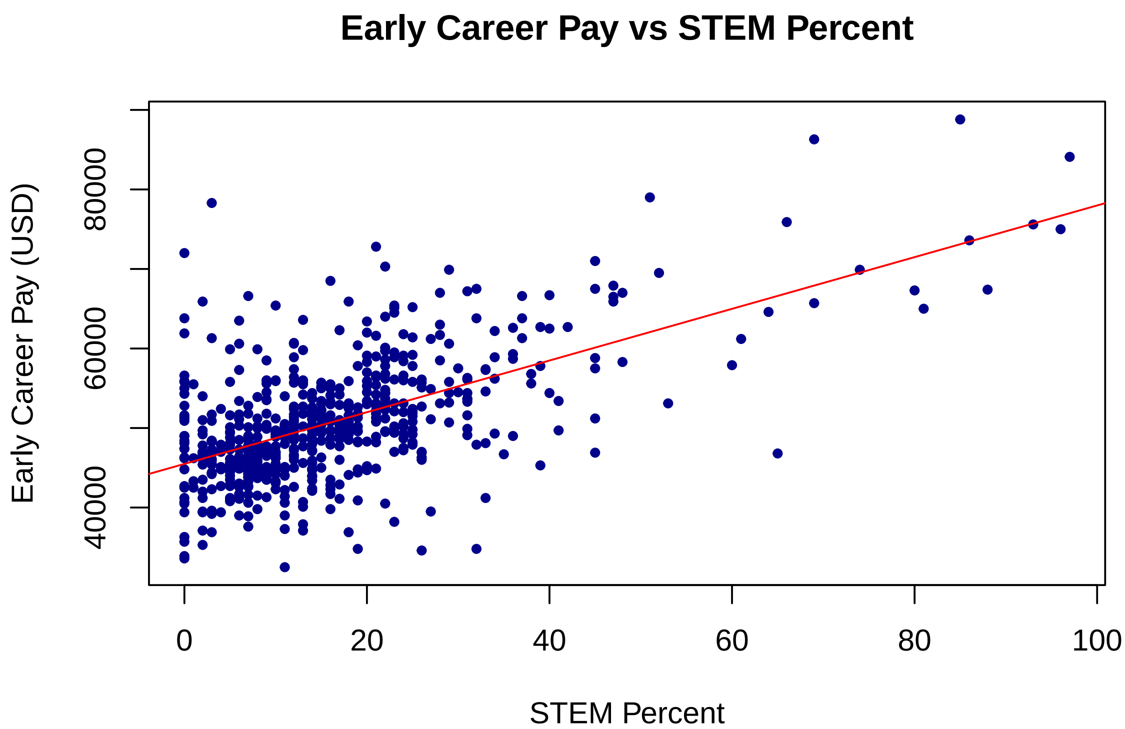
b. Visualize the relationship between (i) `early_career_pay` and `type` and (ii) `early_career_pay` and `stem_percent`. Write an observation from each plot.

```r
boxplot(early_career_pay ~ type, data = college_data,
        main = "Early Career Pay by College Type",
        xlab = "College Type",
        ylab = "Early Career Pay (USD)",
        col = c("lightblue", "lightgreen"))
```




**From this box plot, we can see that the median early career pay is higher for private colleges than it is for public colleges. We can also see that highest early career pay is accounted for a private institution.**

```
plot(college_data$stem_percent, college_data$early_career_pay,
     main = "Early Career Pay vs STEM Percent",
     xlab = "STEM Percent",
     ylab = "Early Career Pay (USD)",
     pch = 19, cex = 0.6, col = "darkblue")
abline(lm(early_career_pay ~ stem_percent, data = college_data), col = "red")
```




**From this plot, we can see that there is no clear relationship between STEM Percent and early career pay when STEM percent is below 40%, however when STEM percent is above 40% we see that there is a positive correlation between STEM Percent and early career pay. So, for values of STEM Percent above 40%, as STEM percent increases, Early Career Pay also increases.**

c. Below is the specification of the statistical model for this analysis. Fit the model and neatly display the results using 3 digits. Display the 95% confidence interval for the coefficients.

$$
\begin{aligned}
early\_career\_pay_i = \beta_0 + \beta_1\ out\_of\_state\_total_i + \beta_2\ type & \quad (1)\\
+ \beta_3\ stem\_percent_i + \beta_4\ type * stem\_percent_i & \quad (2)\\
+ \epsilon_i, \qquad \text{where } \epsilon_i \sim N(0, \sigma^2) & \quad (3)
\end{aligned}
$$

```r
library(broom)
modelearly <- lm(early_career_pay ~ out_of_state_total + type + stem_percent + type:stem_percent,
            data = college_data)
tidy_modelearly <- tidy(modelearly) %>%
  mutate(across(where(is.numeric), ~ round(., 3)))
tidy_modelearly
```

```
## # A tibble: 5 x 5
##   term                       estimate std.error statistic p.value
##   <chr>                         <dbl>     <dbl>     <dbl>   <dbl>
## 1 (Intercept)               36218.       850.        42.6     0
## 2 out_of_state_total            0.253      0.018     13.7     0
## 3 typePublic                 1185.       769.         1.54    0.124
```

```
## 4 stem_percent              214.        19.3       11.1    0
## 5 typePublic:stem_percent    49.5       33.9        1.46   0.144
```

```r
confint(modelearly) %>%
  round(3)
```

```
##                              2.5 %    97.5 %
## (Intercept)              34547.862 37887.546
## out_of_state_total           0.217     0.289
## typePublic                -324.813  2694.853
## stem_percent               176.402   252.211
## typePublic:stem_percent    -16.992   116.069
```

d. How many degrees of freedom are there in the estimate of the regression standard error $\sigma$?

   **Degrees of freedom = number of observations - number of parameters estimated. 593 (observations in the data set) - 5 (parameters estimated in the model (intercept, out_of_state_total , type, stem_percent, type:stem_percent)**

e. What is the 95% confidence interval for the amount in which the intercept for public institutions differs from private institutions?

```r
confint(modelearly)[grep("type", rownames(confint(modelearly))), ]
```

```
##                              2.5 %    97.5 %
## typePublic              -324.81328 2694.8529
## typePublic:stem_percent  -16.99203  116.0687
```

**The difference in intercepts between public and private institutions is represented by**

$$\beta_2 = typePublic$$

**. The 95% confidence interval for this difference is (-324.813, 2694.853).**

### Exercise 6

Use the analysis from the previous exercise to write a paragraph (~ 4 - 5 sentences) describing the differences in early career pay based on the institution characteristics. *The summary should be consistent with the results from the previous exercise, comprehensive, answers the primary analysis question, and tells a cohesive story (e.g., a list of interpretations will not receive full credit).*

**Based on the regression model, early career pay is influenced by both the cost of attendance and characteristics of the institution, including whether a school is public or private and the proportion of STEM degrees awarded. Holding everything else constant, higher out of state total costs are associated with higher early career salaries, suggesting that more expensive institutions tend to produce graduates who earn more sooner after graduation. The interaction between institution type and STEM percentage indicates that the benefit of STEM-heavy programs differs for public versus private schools; specifically, as the STEM share increases, private institutions show a stronger positive association with early career pay than public institutions. While the estimated intercept for public schools is slightly lower than for private schools, the confidence interval for this difference includes zero, implying that there is not strong evidence of a meaningful baseline pay gap between public and private institutions after accounting for costs and STEM emphasis. Overall, the model suggests that graduates from higher cost, STEM focused private institutions tend to earn more early in their careers, but the difference between public and private schools is not statistically significant when these factors are controlled.**

## Grading

| **Total** | **50** |
|---|---|
| Ex 1 | 8 |
| Ex 2 | 4 |
| Ex 3 | 7 |
| Ex 4 | 12 |
| Ex 5 | 12 |
| Ex 6 | 4 |
| Workflow & formatting | 3 |

The "Workflow & formatting" grade is to based on the organization of the assignment write up along with the reproducible workflow. This includes having an organized write up with neat and readable headers, code, and narrative, including properly rendered mathematical notation. It also includes having a reproducible Rmd/Quarto document that can be rendered to reproduce the submitted PDF.